Impairment of endothelial function after a high-fat meal in patients with coronary artery disease.

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OBJECTIVE: To determine the effect of postprandial lipid changes on endothelial function in patients with coronary artery disease (CAD) after a high-fat meal. METHODS: We studied 50 CAD patients and 25 control participants, who were all normocholesterolemic. Flow-mediated vasodilatation of the brachial artery was evaluated by the high-resolution ultrasound technique before and after a single high-fat meal (800 calories; 50 g fat). RESULTS: Postprandial serum triglyceride level increased significantly at 2-7 h and mean flow-mediated vasodilatation was impaired significantly (from 4.22 +/- 0.44 to 2.75 +/- 0.33%, $P < 0.01$) for 75 subjects. The increment in 2 h serum triglyceride level correlated positively with the decrement in postprandial flow-mediated vasodilatation ($r = 0.459$, $P < 0.01$). Postprandial triglyceride level was significantly higher in CAD patients than in control participants. Flow-mediated vasodilatation was significantly impaired in CAD patients (from 3.04 +/- 0.39 to 1.69 +/- 0.23%, $P < 0.01$) and control participants (from 6.58 +/- 0.52 to 4.87 +/- 0.19%, $P < 0.05$) after a high-fat meal. The impairment of flow-mediated dilatation was more severe in CAD patients (44.41%) than in control participants (25.99%, $P < 0.01$). CONCLUSION: Postprandial endothelium-dependent vasodilatation after a single high-fat meal was severely impaired in normocholesterolemic CAD patients and control participants. The disordered postprandial metabolism of triglyceride-rich lipoproteins may play an atherogenic role by inducing endothelial dysfunction.